Market Failure in India's Agricultural Markets

ECMA 35530 Final Presentation

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Introduction

Motivation

- Farmers in India are striking (Indian Express, 2020) for greater public investment in agricultural markets as well as more expenditure on the quality of existing Agriculture Produce Market Committees (APMC) market hubs (‘mandis’).
- Mandis are sparse with concentration in some regions. Their need far outnumbers their numbers.
- We are primarily interested in investigating the effect mandis have on relative wealth – to drive the rationale for addressing farmers’ needs by improving the mandi infrastructure.

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APMC = Agriculture Produce Market Committees

Each state sets up their APMCs to manage the provision of agricultural trade as well as stakeholders (farmers, traders, middlemen) in their mandis. There are 2477 principal regulated markets (mandis) based on geography and 4843 sub-market yards regulated by the respective APMCs in India.

Traders = Consumers in this market while Farmers = Sellers.

Setting and Background

- In India, agriculture is a subject managed by each state.
- Each state sets up their APMCs to manage the provision of agricultural trade and stakeholders (farmers, traders, middlemen/brokers) in their mandis.
- Farmers benefit from mandis in multiple ways.
  - Better prices determined through market equilibrium through an auction process, government regulation of contracts, formalized processes
- Mandis are not growing near as fast as the agricultural output.
  - ! By 2006, the growth rate of mandis declined to less than one-quarter of the growth in crop output
- The poor market infrastructure mean that more produce is sold outside markets than in regulated APMC mandis.

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- The poor market infrastructure mean that more produce is sold outside markets than in regulated APMC mandis.

- Challenge for farmers: market facilities did not keep up with the increase in output, while initial regulation did not allow farmers to sell outside the APMC market $\Rightarrow$ no choice but to resort to middlemen.
- Interlocked transactions $\Rightarrow$ rob farmers of their choice to decide to whom and where to sell.

Motivation

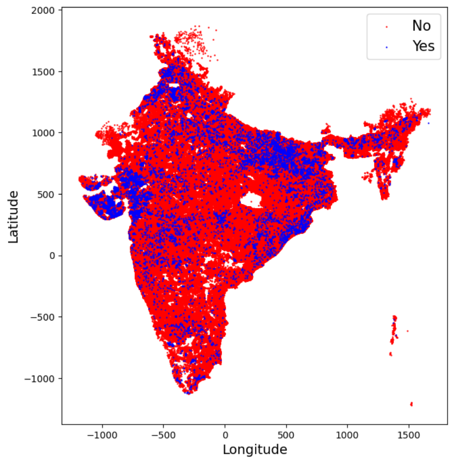

Figure 1: Distribution of Mandi Presence Across India
Source: Authors using SHRUG Data

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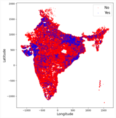

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The spatial trends revealed here map directly onto India's agricultural growth experience – with each state undertaking a radically different trajectory through structural transformation. States that predominantly grow wheat and rice – namely Punjab and Haryana – and have benefitted from the modernization of agriculture have the highest market densities today. Greater levels of commercialization, and thus linkages to markets, also spurred economic development in these states. In contrast, low uptake of productivity-enhancing technologies during the early development stages of the agriculture sector in the northeastern states resulted in low agricultural growth and is reflected in lower market densities in these areas.

Leading into Market Failure

- According to data by NSSO (2021), around 6% of farmers get the government mandated Minimum Support Price (MSP), who sell their produce in state-government regulated mandis to the government or government-adjacent traders (such as some public food processing businesses), and **94% of farmers sell outside mandis or to traders through informal channels (other than the mandi auction system)**.
- Thus, there is exploitation to farmers, with poorer prices outside mandis.
- Mandis are inadequate, in terms of both infrastructure and numbers, but are extremely important for farmer welfare.

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It is important to note here – why do farmers have to go to mandis or necessarily sell through mandis?

- it is the structured market; regulation by APMC staff
- it is the right place to sell to wholesalers/large retailers/traders who are licensed and registered with the APMC
- it ensures the optimum market price through an auction mechanism
- the mandi is regulated by the APMC staff and regulations
- government only regulates – does not intervene on pricing, etc.

Hypotheses & Evidence

Investigation

This intuition motivates a series of questions:

- How does the presence of mandis affect the relative wealth of a region?
  We look at this as a way to get insight into farmer welfare - their ability to access mandis to sell produce and make a living
- How are mandis distributed across India's various districts?
- What kind of insights can we gather, into the mandi system and the associated market failures, to better inform policies that better serve farmers?

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- What kind of insights can we gather, into the mandi system and the associated market failures, to better inform policies that better serve farmers?

- Inefficient or insufficient agricultural markets may lead to sub-optimal pricing and inadequate market access for farmers, thereby reducing the relative wealth of a region.
- Since mandis are determined by local governments, we investigate the possibility that the state governments have done their due diligence and are placing mandis optimally: where there is high production capacity.

Hypotheses

What could be the associated market failures?

1. **Inadequate provision of public goods (public infrastructure)**
   1.1 Existing mandis are not large, well-equipped, and efficient enough to handle all produce and ensure farmers a fair chance of selling
   1.2 Sparse presence/not enough mandis across all regions
2. **Information sparsity**
   2.1 Amongst farmers who are unable to access mandis for fair pricing
      2.1.1 No matter how small transaction costs become, some small farmers will still be unable to make it to mandis, however, information can impart bargaining power $\Rightarrow$ better prices (Aker and Mbiti, 2010)
   2.2 Amongst intermediaries who provide services for both buyers (traders) and farmers
      2.2.1 Intermediaries require resources and information to compete more efficiently to address traders' and farmers' needs
      2.2.2 Information transparency can also address exploitation of farmers (arbitrary prices)

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Transaction costs for farmers can be reduced by:

- simply building more mandis (infeasible, not a sustainable solution)
- having more mandis closeby (not optimal)
- having bigger mandis (not necessarily feasible, but can be explored strategically to ensure better access to mandi benefits for all – yards, hubs, electronic/remote facilities, etc.)

Our analysis uses SHRID level data (created by DDL - somewhere between a village and town level). The outcome variable is 'mean_rwi'. The relative wealth index has a mean value of zero and a standard deviation of one. This standardization does not fundamentally change how we would approach analyzing the impact of 'mandis' on regional units' wealth. It does, however, make it easier to interpret the results since the relative wealth index is already on a comparable scale across different regions.

Evidence - I

Does the average relative wealth differ between parts of districts with mandis and those without?[1]

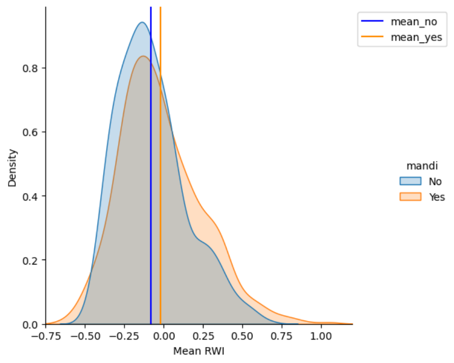
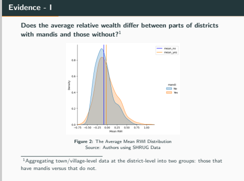


Figure 2: The Average Mean RWI Distribution
Source: Authors using SHRUG Data

[1] Aggregating town/village-level data at the district-level into two groups: those that have mandis versus that do not.



The data is slightly right skewed and is not exactly normally distributed. Thus, we use Mann Whitney U Test (assuming independent samples), or Wilcoxon Signed Rank test (related samples), which are non-parametric variations for the t-test. For this question, we lean towards the use of Wilcoxon Signed Rank Test because SHRID-level within the same district share variables together.

- All three tests (t-test, Mann Whitney or Wilcoxon) indicate a significant difference in RWI between places with mandis and without mandis.
- And when testing with alternative hypothesis that the areas without mandis have less mean RWI, we also reject the null hypothesis. Thus, there is statistically significant evidence that the average RWI is lower in areas without mandis.
- We feel comfortable concluding that the average relative wealth differs between parts of districts with mandis and those without.

Evidence - I

1. The presence of mandis is associated with an increase in the mean relative wealth index by **0.0636 units**, compared to similar regions without mandis - matched through propensity scores computed through a logistic regression, considering covariates related to agricultural indicators (which can affect treatment assignment i.e. presence of mandis).
2. Since the wealth index has a mean of zero and a standard deviation of one, an increase of 0.0636 can be considered a modest positive effect in the context of this scale.

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Steps followed for carrying out the PSM-based causal analysis to compute the ATT:

- Covariates to be included determined
- Logistic regression to obtain a PS for each SHRID
- Match control (no mandi) and treatment (mandi) SHRIDs on the PS using nearest neighbor matching
- Check for balance on treatment and control post matching
- Compute the ATT estimate with this match sample

Evidence - II

Are local governments optimizing by putting mandis where productive land is?

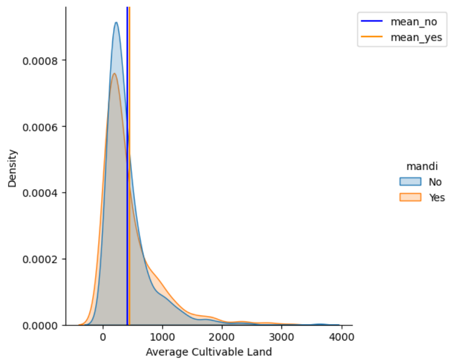

Figure 3: The Average Cultivable Land Distribution
Source: Authors using SHRUG Data

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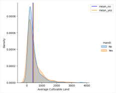


Figure 3: The Average Cultivable Land Distribution
Source: Authors using SHRUG Data

Since the distributions, once again, are right-skewed, and we assume that land within the same district will not be independent, we use the Wilcoxon Signed Rank test again.

We cannot reject the null hypothesis. So there is no statistical evidence that there are significant differences between cultivable land in areas with mandis and areas without mandis.

Thus, we conclude that the local government does NOT seem to be placing mandis where it is more productive/where there is most potential productivity.

Evidence - III

Are local governments optimizing by placing mandis strategically centered within the districts?

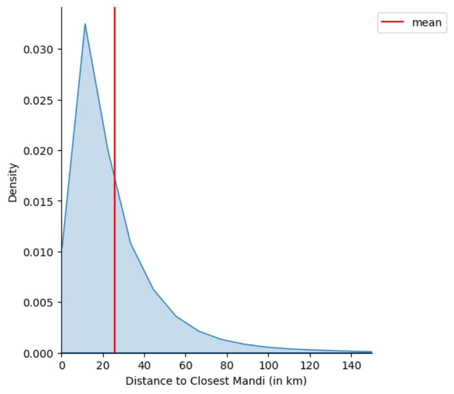
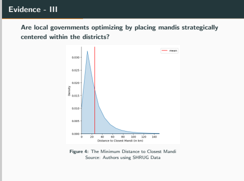


Figure 4: The Minimum Distance to Closest Mandi
Source: Authors using SHRUG Data



In this plot, we have calculated the distances from each SHRID without mandis to every SHRID with mandis within the same district, and then we chose the minimum distance for each SHRID without mandis, assuming that farmers from areas without mandis will go to the closest mandi to them.

- We see that the average distance a farmer has to cross is approximately 27 KMs.
- Calculating the average fuel cost for this distance with some assumptions, it would be on average around 4 dollars one way.
- This is simply fuel cost, no other cost, such as

* possibly renting a truck (only feasible way to cross ~27 kms with produce),
* or opportunity cost of traveling back and forth without guarantees they can fit in the mandi/likely they will not be able to based on qualitative evidence of the small size of mandis.

Policy Recommendations

Policy Recommendations - Inadequate provision of public goods

- Improving mandi infrastructure
  - Better mandi yards, expansion of storage facilities for perishables
  - Increase bargaining power of farmers by reducing urgency of sales of perishable commodities
  - State-regulated private channels (e.g. Bihar, mediates sales and reporting mechanisms)
- Expanding mandi benefits beyond the physical borders
  - access to credit through APMC and central government provisions, reforms to strictly regulate middlemen (commissioned agents), quality assessment of produce, uniform weights
- Increasing number of mandis strategically, while also increasing access to high potential productivity areas through public infrastructure such as roads, transport facilities, and facilitation to help get farmers to the closest Mandi without high costs on the individual farmer

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Enhancing the infrastructure offered by mandis: expanding boundaries, investing in more space and tools, have more streamlined operations:

- Make registration with APMCs as sellers/traders simpler and accessible to all
- This would lower the entry/fixed costs to farmers as well as traders - bringing them both to the common, regulated table
- Conduct is regulated anyways by mandis so that helps the competitiveness of the markets

There exist positive externalities from the provision of public goods like mandis infrastructure

- Allows welfare distribution to a better extent - fairer distribution of marginal benefits, better chances for all types of farmers to sell at optimal rates, no exploitative prices and greater revenue for the state governments

Policy Recommendations - Information sparsity

- **Information sparsity**
  - Regulation; inclusion of middlemen/intermediaries through information dissemination
    - *e-choupals* and hubs set up outside the mandi yards
  - E-mandis (e.g. E-National Agricultural Market)
    - Improve access, scope, and expansion - e.g. Karnataka
    - Only 14% farmers enrolled at the national level (Chaudhary and Suri, 2022)
  - Text messages or other mobile-based methods for information dissemination about pricing/markets
    - 25% farmers use mobile phones to access information about agriculture and livestock, 23% to buy and sell products, and 18% to receive news updates (Kumar, 2023)
    - Mobile density of 82.54% across the nation (Telecom Regulatory Authority of India, 2024)

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- *e-choupal* tackles weak infrastructure and the involvement of intermediaries by installing computers with Internet access in rural areas of India to offer farmers up-to-date marketing and agricultural information. Was implemented by a private company (ITC) that procures from farmers.
- National Agriculture Market or eNAM is an online trading platform for agricultural commodities in India initiated by the Ministry of Agriculture.

Appendix

Data

- **Main data source:** Development Data Lab's SHRUG Datasets (Asher et al., 2021, Chi et al., 2022, Ministry of Rural Development, 2020a, Ministry of Rural Development, 2020b)
- **Datasets used:**
  - Mission Antyodaya Village Facilities (2020): Data on village facilities from the Ministry of Rural Development
  - Facebook wealth and population (2021): Facebook Relative Wealth Relative wealth data from Facebook
  - Shrug Location Names and Additional Keys: SHRUG keys for linking identifiers across SHRUG datasets, location name strings, as well as SHRID spatial statistics
  - SECC Rural: Rural data from the 2011 Socio-Economic and Caste Census (assets, consumption, poverty rates, household size, and agriculture variables)
- **Relative Wealth Index (RWI) variable**:
  - Facebook relative wealth index (RWI) predicted for every populated 2.4-km grid cell in India. It from a supervised ML model from Demographic and Health Surveys spatially linked to features constructed from various sources + aggregated & deidentified connectivity data from Facebook. The data was release in 2021 and uses the DHS data round 2015-16 for India.

Appendix

Analysis - I

In order to match on district-level areas/SHRIDs with mandis vs. not, we first

- Divide into two dataframes, one that has all instances of districts on SHRID level that have a mandi, and the other is for the ones that do not have a mandi.
- Then we will query each dataframe by district name using df.query(district)['facebook_mean_rwi'].sum()
- Get total occurrences of this district in the dataframe and take the average of the RWI of the district, aggregated from SHRID level.

Analysis - II

- For PSM: The presence of mandis is associated with an increase in the mean relative wealth index by 0.0636 units, compared to similar regions without mandis - matched through propensity scores computed through a logistic regression, considering covariates related to agricultural indicators (which can affect treatment assignment i.e. presence of mandis). Since the wealth index has a mean of zero and a standard deviation of one, an increase of 0.0636 can be considered a modest positive effect in the context of this scale.
- For geospatial analysis: used SHRID location data and district information, converted to latitudes and longitudes, used Python libraries to plot color-coded locations on the map

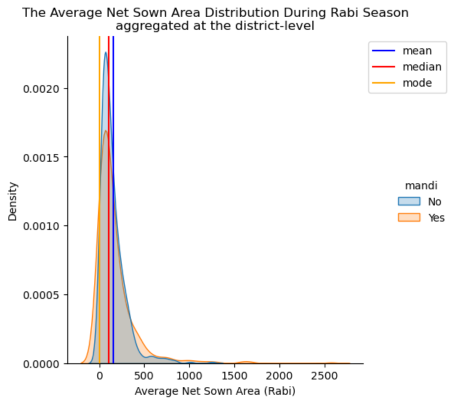
Analysis - III

Limitations of this causal analysis:

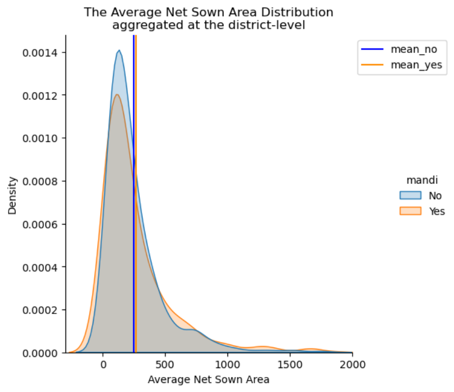
- reliance on observational data that may be susceptible to measurement errors
- we basically assumed away the fact that that there are no unobserved confounders affecting both the presence of mandis and the wealth index
- we have been limited by the quality/efficiency of the covariates we have used (omitted variable bias remains a possibility)
- there might be a common support problem only performed NN – it might not be the correct algorithm for this case

Analysis - IV(i)

Measuring productive capacity in terms of Net Sown Area and Net Sown Area during Rabi season



Analysis - IV(ii)



Analysis - V

Test 1: Average RWI in SHRID w/ mandis VS w/o mandis

```python
# still not fully normal but not very skewed either?

# ttest
print(stats.ttest_ind(lst_avg_no_mandi, lst_avg_yes_mandi))
print(stats.ttest_rel(lst_avg_no_mandi, lst_avg_yes_mandi))
# MannWhitney Test; assuming indepndent samples
print(stats.mannwhitneyu(lst_avg_no_mandi, lst_avg_yes_mandi)) #Probably more reliable
print(stats.mannwhitneyu(lst_avg_no_mandi, lst_avg_yes_mandi, alternative='less')) #Probably more reliable

# Wilcoxon; assuming related samples
print(stats.wilcoxon(lst_avg_no_mandi, lst_avg_yes_mandi))
print(stats.wilcoxon(lst_avg_no_mandi, lst_avg_yes_mandi, alternative='less'))
```

✓ 0.0s

```
TtestResult(statistic=-4.102651303088297, pvalue=4.3851872740044774e-05, df=1102.0)
TtestResult(statistic=-9.761525164478435, pvalue=7.201562643462389e-21, df=551)
MannwhitneyuResult(statistic=131848.0, pvalue=0.00010848798324749302)
MannwhitneyuResult(statistic=131848.0, pvalue=5.424399162374651e-05)
WilcoxonResult(statistic=38481.0, pvalue=6.013590992406874e-24)
WilcoxonResult(statistic=38481.0, pvalue=3.006795496203437e-24)
```

Test 2: Average Cultivable Land in SHRID w/ mandis VS w/o mandis

Average Cultivable Land

```python
print(stats.mannwhitneyu(lst_pprod2_no, lst_pprod2_yes)) # Assuming independence; does not work. They are geographically within same area, so..
print(stats.wilcoxon(lst_pprod2_no, lst_pprod2_yes)) # Assuming they are related; The Wilcoxon signed-rank statistic.
```

[169] ✓ 0.0s Python

```
MannwhitneyuResult(statistic=157333.0, pvalue=0.34708932729766584)
WilcoxonResult(statistic=74215.0, pvalue=0.5755526188232595)
```

Test 3: Net Sown Area w/ mandis VS w/o mandis

```
print('\n')
print(stats.ttest_rel(lst_cprod1_no, lst_cprod1_yes))
print(stats.ttest_rel(lst_cprod1_no, lst_cprod1_yes, alternative='less'))
print('\n')
print(stats.wilcoxon(lst_cprod1_no, lst_cprod1_yes))
print(stats.wilcoxon(lst_cprod1_no, lst_cprod1_yes, alternative = 'less'))
```

[196] ✓ 0.0s

```
...

TtestResult(statistic=-2.4007165174929708, pvalue=0.016693542835929012, df=551)
TtestResult(statistic=-2.4007165174929708, pvalue=0.008346771417964506, df=551)


WilcoxonResult(statistic=75470.0, pvalue=0.8218775575615409)
WilcoxonResult(statistic=75470.0, pvalue=0.41093877878077045)
```

Test 4: Net Sown Area during Rabi season w/ mandis VS w/o mandis

```
# ASSUMING INDEPENDENCE DOES NOT MAKE SENSE B/C IT IS WITHIN SAME DISTRICT
# print(stats.ttest_ind(lst_cprod1_no, lst_cprod1_yes))
# print(stats.ttest_ind(lst_cprod1_no, lst_cprod1_yes, alternative = 'greater'))
# print(stats.ttest_ind(lst_cprod1_no, lst_cprod1_yes, alternative = 'less'))
print('\n')
print(stats.ttest_rel(lst_cprod1_no, lst_cprod1_yes))
print(stats.ttest_rel(lst_cprod1_no, lst_cprod1_yes, alternative='less'))
print('\n')
print(stats.wilcoxon(lst_cprod1_no, lst_cprod1_yes))
print(stats.wilcoxon(lst_cprod1_no, lst_cprod1_yes, alternative = 'less'))

# print(stats.mannwhitneyu(lst_cprod1_no, lst_cprod1_yes))
# print(stats.mannwhitneyu(lst_cprod1_no, lst_cprod1_yes, alternative='greater'))
#print(stats.mannwhitneyu(lst_cprod1_no, lst_cprod1_yes, alternative='less'))
```

[166] ✓ 0.0s

```
...

TtestResult(statistic=-2.4007165174929708, pvalue=0.016693542835929012, df=551)
TtestResult(statistic=-2.4007165174929708, pvalue=0.008346771417964506, df=551)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```

Auxiliary Analysis - I

Regression 1: Initial Specification

OLS Regression Results

Dep. Variable:	facebook_mean_rwi	**R-squared:**	0.001
Model:	OLS	**Adj. R-squared:**	0.001
Method:	Least Squares	**F-statistic:**	532.6
Date:	Sun, 25 Feb 2024	**Prob (F-statistic):**	8.75e-118
Time:	23:48:40	**Log-Likelihood:**	-1.5052e+05
No. Observations:	501446	**AIC:**	3.010e+05
Df Residuals:	501444	**BIC:**	3.011e+05
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	-0.1143	0.000	-244.853	0.000	-0.115	-0.113
mandi	0.0710	0.003	23.079	0.000	0.065	0.077

Omnibus:	24954.262	**Durbin-Watson:**	0.589
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	28913.292
Skew:	0.568	**Prob(JB):**	0.00
Kurtosis:	3.305	**Cond. No.**	6.67

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Auxiliary Analysis - II

Regression 2: Govt. Agri Schemes Covs. Added

OLS Regression Results

Dep. Variable:	facebook_mean_rwi	**R-squared:**	0.015
Model:	OLS	**Adj. R-squared:**	0.015
Method:	Least Squares	**F-statistic:**	1259.
Date:	Fri, 16 Feb 2024	**Prob (F-statistic):**	0.00
Time:	21:33:44	**Log-Likelihood:**	-1.4704e+05
No. Observations:	501446	**AIC:**	2.941e+05
Df Residuals:	501439	**BIC:**	2.942e+05
Df Model:	6		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	-0.1459	0.001	-216.045	0.000	-0.147	-0.145
mandi	0.0619	0.003	20.222	0.000	0.056	0.068
total_hhd_availing_pmuy_benefits	-6.795e-05	3.7e-06	-18.351	0.000	-7.52e-05	-6.07e-05
total_hhd_availing_pmjdy_bank_ac	8.135e-05	2.99e-06	27.163	0.000	7.55e-05	8.72e-05
total_hhd_having_bpl_cards	8.941e-05	2.37e-06	37.748	0.000	8.48e-05	9.41e-05
is_pds_available	0.0392	0.001	40.982	0.000	0.037	0.041
total_hhd_having_pmsbhgy_benefit	-9.623e-05	4.13e-06	-23.279	0.000	-0.000	-8.81e-05

Omnibus:	26437.908	**Durbin-Watson:**	0.605
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	30933.928
Skew:	0.584	**Prob(JB):**	0.00
Kurtosis:	3.340	**Cond. No.**	2.60e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 2.6e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Auxiliary Analysis - III

Regression 3: Govt. Schemes Covs. Added

OLS Regression Results

Dep. Variable:	facebook_mean_rwi	**R-squared:**	0.014
Model:	OLS	**Adj. R-squared:**	0.014
Method:	Least Squares	**F-statistic:**	1039.
Date:	Fri, 16 Feb 2024	**Prob (F-statistic):**	0.00
Time:	21:33:50	**Log-Likelihood:**	-1.4717e+05
No. Observations:	501446	**AIC:**	2.944e+05
Df Residuals:	501438	**BIC:**	2.944e+05
Df Model:	7		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	-0.1416	0.001	-236.856	0.000	-0.143	-0.140
mandi	0.0514	0.003	16.723	0.000	0.045	0.057
sub_centre	0.0274	0.001	20.260	0.000	0.025	0.030
phc	0.0385	0.002	16.168	0.000	0.034	0.043
chc	0.0543	0.002	31.197	0.000	0.051	0.058
availability_of_mother_child_hea	0.0325	0.001	31.746	0.000	0.031	0.035
total_hhd_registered_under_pmjay	0.0001	3.15e-06	36.202	0.000	0.000	0.000
gp_total_no_of_beneficiaries_rec	3.144e-05	4.64e-06	6.775	0.000	2.23e-05	4.05e-05

Omnibus:	26163.492	**Durbin-Watson:**	0.601
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	30553.138
Skew:	0.581	**Prob(JB):**	0.00
Kurtosis:	3.332	**Cond. No.**	1.35e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.35e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

Auxiliary Analysis - IV

Regression 4: Allied Agri. Covs. Added

OLS Regression Results

Dep. Variable:	facebook_mean_rwi	**R-squared:**	0.015
Model:	OLS	**Adj. R-squared:**	0.015
Method:	Least Squares	**F-statistic:**	827.9
Date:	Fri, 16 Feb 2024	**Prob (F-statistic):**	0.00
Time:	21:48:35	**Log-Likelihood:**	-1.4709e+05
No. Observations:	501446	**AIC:**	2.942e+05
Df Residuals:	501436	**BIC:**	2.943e+05
Df Model:	9		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.1277	0.001	-217.576	0.000	-0.129	-0.127
mandi	0.0523	0.003	16.996	0.000	0.046	0.058
availability_of_poultry_dev_proj	0.0827	0.002	42.224	0.000	0.079	0.087
availability_of_goatary_dev_proj	-0.0131	0.002	-6.536	0.000	-0.017	-0.009
availability_of_pigery_developme	-0.0623	0.003	-24.089	0.000	-0.067	-0.057
fpo	0.0332	0.001	26.761	0.000	0.031	0.036
availability_of_food_storage_war	-0.0050	0.002	-2.626	0.009	-0.009	-0.001
availability_of_fish_community_p	0.0117	0.001	9.474	0.000	0.009	0.014
livestock_ext_services	-0.0316	0.001	-27.980	0.000	-0.034	-0.029
is_veterinary_hospital_available	0.0781	0.002	47.677	0.000	0.075	0.081

Omnibus:	25690.678	**Durbin-Watson:**	0.609
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	29941.564
Skew:	0.574	**Prob(JB):**	0.00
Kurtosis:	3.336	**Cond. No.**	7.48

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Auxiliary Analysis - V

Regression 5: Agricultural Covs. Added

OLS Regression Results

Dep. Variable:	facebook_mean_rwi	**R-squared:**	0.030
Model:	OLS	**Adj. R-squared:**	0.030
Method:	Least Squares	**F-statistic:**	1554.
Date:	Fri, 16 Feb 2024	**Prob (F-statistic):**	0.00
Time:	21:41:05	**Log-Likelihood:**	-1.4313e+05
No. Observations:	501445	**AIC:**	2.863e+05
Df Residuals:	501434	**BIC:**	2.864e+05
Df Model:	10		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	-0.1485	0.001	-143.349	0.000	-0.150	-0.146
mandi	0.0539	0.003	17.667	0.000	0.048	0.060
total_no_of_farmers	4.71e-05	1.12e-06	42.115	0.000	4.49e-05	4.93e-05
net_sown_area_in_hac	-7.503e-05	1.64e-06	-45.633	0.000	-7.83e-05	-7.18e-05
canal	0.0830	0.002	53.303	0.000	0.080	0.086
surface_water	-0.0095	0.002	-5.388	0.000	-0.013	-0.006
ground_water	0.0363	0.001	30.484	0.000	0.034	0.039
is_fertilizer_shop_available	0.0953	0.001	67.780	0.000	0.093	0.098
area_irrigated_in_hac	1.344e-05	2.3e-06	5.840	0.000	8.93e-06	1.8e-05
is_soil_testing_centre_available	-0.0134	0.003	-5.069	0.000	-0.019	-0.008
is_common_pastures_available	-0.0435	0.001	-39.412	0.000	-0.046	-0.041

Omnibus:	26985.901	**Durbin-Watson:**	0.631
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	31843.101
Skew:	0.584	**Prob(JB):**	0.00
Kurtosis:	3.402	**Cond. No.**	3.97e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.97e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

Auxiliary Analysis - VI

Regression 6: Connectivity Covs. Added

OLS Regression Results

Dep. Variable:	facebook_mean_rwi	**R-squared:**	0.014
Model:	OLS	**Adj. R-squared:**	0.014
Method:	Least Squares	**F-statistic:**	778.6
Date:	Fri, 16 Feb 2024	**Prob (F-statistic):**	0.00
Time:	21:48:59	**Log-Likelihood:**	-1.4730e+05
No. Observations:	501446	**AIC:**	2.946e+05
Df Residuals:	501436	**BIC:**	2.947e+05
Df Model:	9		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.1543	0.001	-184.501	0.000	-0.156	-0.153
mandi	0.0454	0.003	14.732	0.000	0.039	0.051
no_of_physically_challenged_recv	0.0002	1.91e-05	9.521	0.000	0.000	0.000
internal_pucca_road	0.0024	0.001	2.575	0.010	0.001	0.004
is_community_biogas_waste_recycl	-0.0046	0.002	-2.249	0.025	-0.009	-0.001
recreational_centre	0.0300	0.001	27.527	0.000	0.028	0.032
availability_of_custom_hiring_ce	0.0144	0.002	7.570	0.000	0.011	0.018
csc	0.0164	0.001	14.981	0.000	0.014	0.019
is_pds_available	0.0328	0.001	33.488	0.000	0.031	0.035
is_post_office_available	0.0414	0.001	33.958	0.000	0.039	0.044

Omnibus:	26604.341	**Durbin-Watson:**	0.602
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	31153.282
Skew:	0.586	**Prob(JB):**	0.00
Kurtosis:	3.340	**Cond. No.**	164.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Auxiliary Analysis - VII

Regression 7: Complete Specification

	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.1504	0.001	-103.281	0.000	-0.153	-0.148
mandi	0.0369	0.003	12.190	0.000	0.031	0.043
no_of_physically_challenged_recv	8.422e-05	1.89e-05	4.457	0.000	4.72e-05	0.000
internal_pucca_road	-0.0006	0.001	-0.663	0.507	-0.002	0.001
is_community_biogas_waste_recycl	-0.0025	0.002	-1.225	0.221	-0.007	0.002
recreational_centre	0.0270	0.001	24.514	0.000	0.025	0.029
availability_of_custom_hiring_ce	0.0031	0.002	1.538	0.124	-0.001	0.007
csc	0.0127	0.001	11.348	0.000	0.010	0.015
is_pds_available	0.0144	0.001	28.766	0.000	0.013	0.015
is_post_office_available	0.0124	0.001	9.286	0.000	0.010	0.015
total_no_of_farmers	5.907e-06	1.28e-06	4.609	0.000	3.4e-06	8.42e-06
net_sown_area_in_hac	-9.169e-05	1.65e-06	-55.623	0.000	-9.49e-05	-8.85e-05
canal	0.0780	0.002	50.332	0.000	0.075	0.081
surface_water	-0.0129	0.002	-7.394	0.000	-0.016	-0.009
ground_water	0.0384	0.001	32.171	0.000	0.036	0.041
is_fertilizer_shop_available	0.0499	0.002	33.169	0.000	0.047	0.053
area_irrigated_in_hac	7.997e-06	2.28e-06	3.512	0.000	3.53e-06	1.25e-05
is_soil_testing_centre_available	-0.0357	0.003	-12.831	0.000	-0.041	-0.030
is_common_pastures_available	-0.0521	0.001	-46.167	0.000	-0.054	-0.050
availability_of_poultry_dev_proj	0.0600	0.002	31.133	0.000	0.056	0.064
availability_of_goatary_dev_proj	-0.0109	0.002	-5.553	0.000	-0.015	-0.007
availability_of_pigery_developme	-0.0473	0.003	-18.496	0.000	-0.052	-0.042
fpo	0.0142	0.001	11.280	0.000	0.012	0.017
availability_of_food_storage_war	-0.0188	0.002	-9.524	0.000	-0.023	-0.015
availability_of_fish_community_p	-0.0028	0.001	-2.296	0.022	-0.005	-0.000
livestock_ext_services	-0.0384	0.001	-33.928	0.000	-0.041	-0.036
is_veterinary_hospital_available	0.0311	0.002	17.736	0.000	0.028	0.035
sub_centre	0.0067	0.001	4.796	0.000	0.004	0.009
phc	0.0120	0.002	5.065	0.000	0.007	0.017
chc	0.0256	0.002	14.243	0.000	0.022	0.029
availability_of_mother_child_hea	0.0247	0.001	23.279	0.000	0.023	0.027
total_hhd_registered_under_pmjay	4.473e-05	3.49e-06	12.828	0.000	3.79e-05	5.16e-05
gp_total_no_of_beneficiaries_rec	2.68e-06	4.61e-06	0.581	0.561	-6.36e-06	1.17e-05
total_hhd_availing_pmuy_benefits	-6.939e-05	3.65e-06	-18.995	0.000	-7.65e-05	-6.22e-05
total_hhd_availing_pmjdy_bank_ac	3.316e-05	3.13e-06	10.577	0.000	2.7e-05	3.93e-05
total_hhd_having_bpl_cards	3.933e-05	2.58e-06	15.235	0.000	3.43e-05	4.44e-05
is_pds_available	0.0144	0.001	28.766	0.000	0.013	0.015
total_hhd_having_pmsbhgy_benefit	-8.44e-05	4.05e-06	-20.832	0.000	-9.23e-05	-7.65e-05
total_grd_and_pg_in_village	0.0001	2.32e-06	48.908	0.000	0.000	0.000
availability_of_middle_school	-0.0116	0.001	-11.103	0.000	-0.014	-0.010
availability_of_primary_school	-0.0458	0.001	-34.456	0.000	-0.048	-0.043
availability_of_govt_degree_coll	0.0376	0.003	12.978	0.000	0.032	0.043
availability_of_adult_edu_centre	-0.0358	0.002	-18.621	0.000	-0.040	-0.032
availability_of_ssc_school	0.0393	0.002	23.524	0.000	0.036	0.043
is_aanganwadi_centre_available	0.0192	0.001	14.131	0.000	0.017	0.022
is_vocational_edu_centre_availab	0.0409	0.003	15.739	0.000	0.036	0.046

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